Unit - 2

Double Integral

anton bb
stewart → books.

Iterated integral

$R = [a,b] \times [c,d]$

$$\iint_R f(x,y)\,dx\,dy = \int_c^d \left(\int_a^b f(x,y)\,dx \right) dy$$

Theorem

Let R be the rectangle defined by the inequalities
$a \le x \le b$, $c \le y \le d$

if $f(x,y)$ is continous on this rectangle, then

$$\iint_R f(x,y)\,dA = \int_c^d \int_a^b f(x,y)\,dx\,dy = \int_a^b \int_c^d f(x,y)\,dy\,dx$$

Ex 1 $\iint_R (x\sin y - y\sin x)\,dA$

$R = \{(x,y) \mid 0 \le x \le \pi/2,\ 0 \le y \le \pi/3$

$$\int_0^{\pi/3} \left(\int_0^{\pi/2} x\sin y - y\sin x\,dx \right) dy$$

$$= \int_0^{\pi/3} \left[\frac{x^2}{2}\sin y + y\cos x \right]_0^{\pi/2} dy$$

$$= \int_0^{\pi/3} \left[\frac{\pi^2}{8}\sin y - y \right] dy = \left[\frac{-\pi^2\cos y}{8} - \frac{y^2}{2} \right]_0^{\pi/3}$$

$$= \frac{-\pi^2}{16} - \frac{\pi^2}{18} + \frac{\pi^2}{8}$$

$$= \frac{-9\pi^2 - 8\pi^2 + 18\pi^2}{144} = \left(\frac{\pi}{12}\right)^2$$